

UTI — Diagnosis, Asymptomatic Bacteriuria & Special Hosts (Bazaar)



1. Nitrite

WHAT YOU SEE

Vendor sign 'NITRITE' over dipstick cups

WHAT IT ENCODES

Urine nitrite is produced when Enterobacteriaceae (E. coli, Klebsiella, Proteus) reduce dietary nitrate to nitrite. A POSITIVE nitrite is fairly specific for a gram-negative UTI. It requires adequate bladder dwell time, so a first-morning specimen is best; very dilute or briefly-held urine can read falsely negative.

BOARD TRAP

Nitrite is specific but NOT sensitive — a negative nitrite does not exclude UTI, and organisms that don't reduce nitrate (Enterococcus, Staph saprophyticus, Pseudomonas) give negative nitrite despite true infection.

2. Leukocyte esterase

WHAT YOU SEE

'LE' sign on dipstick stall

WHAT IT ENCODES

Leukocyte esterase detects pyuria (WBCs/neutrophils in urine) and is the sensitive partner to nitrite. LE positive indicates inflammation/pyuria. Combined dipstick: nitrite + LE both positive strongly supports UTI; both negative makes it unlikely. Microscopic pyuria is >10 WBC/hpf.

BOARD TRAP

Sterile pyuria (LE+/pyuria but negative routine culture) should trigger thought of Chlamydia/gonorrhea urethritis, TB, or interstitial nephritis — not just 'treat as UTI.'

3. Dilute = false negative

WHAT YOU SEE

Banner 'DILUTE = FALSE NEG' under nitrite stall

WHAT IT ENCODES

Very dilute urine (high fluid intake, low specific gravity) or short bladder incubation lowers nitrite concentration below the detection threshold, producing false-negative dipstick results. This is why dipstick is a screen, not a rule-out, and clinical symptoms plus culture guide management.

BOARD TRAP

A negative dipstick in a symptomatic patient does not rule out UTI — dilution and non-nitrate-reducers cause false negatives; send a culture when suspicion is high.

4. E. coli — #1 uropathogen

WHAT YOU SEE

'E. COLI ' check on dipstick cup

WHAT IT ENCODES

E. coli causes ~75–95% of uncomplicated UTIs — the dominant uropathogen across cystitis and pyelonephritis. Uropathogenic E. coli uses P-fimbriae/adhesins to ascend. Knowing E. coli is #1 anchors empiric therapy choices (nitrofurantoin, TMP-SMX, fosfomycin) before culture returns.

BOARD TRAP

E. coli is #1 overall, but in a young sexually active woman with nitrite-NEGATIVE UTI think Staphylococcus saprophyticus (the #2 cause in this group); Proteus suggests stones/alkaline urea-splitting infection.

5. Staph saprophyticus

WHAT YOU SEE

'STAPH SAPRO' label at base of stall

WHAT IT ENCODES

Staphylococcus saprophyticus is the second most common cause of acute uncomplicated cystitis in young, sexually active women. It is coagulase-negative, novobiocin-RESISTANT, and is nitrite-negative (doesn't reduce nitrate), so dipstick may mislead.

BOARD TRAP

A young woman with cystitis but NEGATIVE nitrite — don't dismiss UTI. Staph saprophyticus is nitrite-negative; coagulase-negative + novobiocin resistance distinguishes it from S. epidermidis (novobiocin-sensitive).

6. Asymptomatic bacteriuria — TREAT ONLY arch

WHAT YOU SEE

'TREAT ONLY' arch with red X / 'NO TREAT' figures

WHAT IT ENCODES

Asymptomatic bacteriuria (positive culture, NO symptoms) is generally NOT treated — treatment doesn't help and breeds resistance. The major EXCEPTIONS to treat are: (1) pregnancy and (2) before urologic procedures with expected mucosal bleeding (e.g., TURP). The 'TREAT ONLY' arch flags these two carve-outs.

BOARD TRAP

Do NOT treat asymptomatic bacteriuria in the elderly, diabetics, catheterized patients, or pre-non-urologic surgery — only pregnancy and invasive urologic procedures justify treatment. Treating others is a classic wrong answer.

7. Pyuria alone ≠ treat

WHAT YOU SEE

'PYU' figure carrying staff between arches

WHAT IT ENCODES

Pyuria frequently accompanies asymptomatic bacteriuria, particularly in elderly and catheterized patients, and by itself is NOT an indication to treat. Decisions hinge on SYMPTOMS, not on pyuria or a positive culture in an asymptomatic host.

BOARD TRAP

Pyuria in an asymptomatic catheterized or elderly patient is not a reason for antibiotics — pyuria does not upgrade asymptomatic bacteriuria into a treatable UTI.

8. Pregnancy — treat asymptomatic bacteriuria

WHAT YOU SEE

Pregnant woman in green at center of bazaar

WHAT IT ENCODES

In pregnancy, asymptomatic bacteriuria IS screened for and treated because untreated it progresses to pyelonephritis (~30–40%) and is linked to preterm birth and low birth weight. Safe agents: nitrofurantoin (avoid at term/G6PD), amoxicillin, cephalexin, fosfomycin. Avoid fluoroquinolones and TMP-SMX (esp. 1st and 3rd trimesters).

BOARD TRAP

Pregnancy is THE classic exception — always treat asymptomatic bacteriuria in pregnancy. And avoid TMP-SMX near term (kernicterus) and fluoroquinolones throughout.

9. Symptomatic cystitis

WHAT YOU SEE

'SYMPTOMATIC' patient on stretcher, fire behind

WHAT IT ENCODES

Symptomatic UTI = dysuria, frequency, urgency, suprapubic pain (cystitis), or fever/flank pain/CVA tenderness (pyelonephritis). Symptoms — not culture colony counts alone — drive the decision to treat. Acute uncomplicated cystitis in a healthy non-pregnant woman can be treated empirically.

BOARD TRAP

Treat based on symptoms + urinalysis, not on a positive culture in isolation. Fever and flank pain push you from simple cystitis (short oral course) to pyelonephritis (longer, broader therapy).

10. Catheter / transplant host

WHAT YOU SEE

'TRANSPLANT' figure with boxing glove near stretcher

WHAT IT ENCODES

Catheter-associated bacteriuria is usually NOT treated unless symptomatic; replace/remove the catheter first ('replace first'). Renal-transplant and other immunosuppressed hosts are an important nuance — some guidelines screen/treat early post-transplant — but routine asymptomatic catheter bacteriuria is still generally left alone.

BOARD TRAP

For catheter-associated bacteriuria, REPLACE the catheter and treat only if symptomatic — don't reflexively give antibiotics for a positive catheter-urine culture in an asymptomatic patient.

11. Replace catheter first / 7–14 d

WHAT YOU SEE

Right arch 'REPLACE FIRST', '7–14 D' bedside

WHAT IT ENCODES

For catheter-associated UTI that IS symptomatic, exchange the indwelling catheter before/at the start of therapy to remove the biofilm reservoir, then treat 7–14 days depending on severity and response. Source control (catheter exchange/removal) is as important as antibiotic choice.

BOARD TRAP

Antibiotics without removing/exchanging the colonized catheter leads to relapse — biofilm shields organisms. Always pair therapy with catheter exchange or removal.

12. Fluconazole — candiduria

WHAT YOU SEE

'FLUC' bottle on right arch shelf

WHAT IT ENCODES

Candiduria in an asymptomatic patient is usually managed by removing the catheter (often clears spontaneously) — not by antifungals. Treat (fluconazole) only if symptomatic, neutropenic, undergoing urologic manipulation, or in a renal-transplant recipient. Fluconazole is first-line for susceptible Candida albicans urinary infection.

BOARD TRAP

Asymptomatic candiduria generally needs catheter removal, NOT fluconazole — antifungals are reserved for symptomatic disease or high-risk hosts (neutropenia, urologic surgery, transplant).

13. Candida / fungus

WHAT YOU SEE

'CANDIDA / FUNGUS' sign on castle wall

WHAT IT ENCODES

Funguria is usually Candida, seen in catheterized, diabetic, broad-spectrum-antibiotic-exposed, or immunosuppressed patients. Most represents colonization. Pyelonephritis or fungus ball (bezoar) is the concerning invasive form, especially in diabetics, and can cause obstruction.

BOARD TRAP

Most candiduria is colonization, not invasive infection — the leap to systemic antifungals or amphotericin bladder irrigation is rarely warranted and is a classic over-treatment trap.

14. Emphysematous pyelonephritis

WHAT YOU SEE

Right: obese man clutching flank, green gas, 'EMPHYSEMATOUS'

WHAT IT ENCODES

Emphysematous pyelonephritis is a necrotizing, gas-forming renal infection (CT shows gas in the renal parenchyma), occurring almost exclusively in DIABETICS, usually from E. coli or Klebsiella. It is life-threatening: management is aggressive — IV antibiotics + percutaneous drainage, and nephrectomy if it fails.

BOARD TRAP

Gas in the kidney on imaging in a diabetic = emphysematous pyelonephritis, a surgical/drainage emergency — not a simple pyelonephritis you can treat with oral antibiotics. E. coli is the usual culprit.